

Clotilde Cantini

Contact Information

Address: 3 rue Marguerite de Rochechouart, 75009, Paris
Phone: +33 (0)7 83 20 14 00
Email: clotildecantini1@outlook.fr
LinkedIn: <https://www.linkedin.com/in/clotilde-cantini/>
GitHub: <https://github.com/clotildecantini>
Google Scholar : <https://scholar.google.com/citations?user=0bvVQQQAAAAJ&hl=fr>

Scientific Publications and Patents

- **Cantini C**, Rolland-Piegue E, Schmitter D. Exact Implementation of Closed-Form Liquid Neural Networks With Arbitrary Precision. IEEE Signal Processing Letters. 2025 Feb 6.
- Nguyen TD, Lesage A, **Cantini C**, Riad R. Quantized Approximate Signal Processing (QASP): Towards Homomorphic Encryption for audio. arXiv preprint arXiv:2505.10500. 2025 May 15. (submitted – under review)
- T. D. Nguyen, A. Lesage, **C. Cantini**, and R. Riad, “Computer-implemented method for determining a frequency content of an encrypted raw audio signal,” U.S. Patent 12,425,184 B1, Sep. 23, 2025.

Education

Master in Operational Research (MPRO)

September 2024 - April 2025

Ensta ParisTech, Paris, France

MSc in Operational Research and Optimization

Relevant Courses: Optimization and graphs, Metaheuristics, Convex Optimization, Deep Learning for Combinatorial Optimization Problems, Advanced Combinatorial Optimization, Constraints programming, Graphs.

Engineering School

September 2021 - June 2023

École des Ponts ParisTech, Paris, France

MSc in Mathematical and Computer Engineering

Relevant Courses: Software Engineering, Deep Learning, Machine Learning, Convex Optimization, Information Processing and Machine Learning, Advanced Programming and Algorithms, Operational Research, Stochastic Processes and Applications.

Preparatory Class, MPSI/PSI*

2019 - 2021

Lycée Montaigne, Bordeaux, France

Two-year intensive maths/physics program preparing for the national competitive exam for entry to engineering schools

Baccalaureate

2019

Lycée Beaulieu, Cognac, France

Scientific baccalaureate (*highest honors*)

Work Experience

Pre-PhD research assistant

May 2026 - December 2026

CERMICS, École des Ponts ParisTech, France

- Research Area: Airline Revenue Management
- Specialization: Competition-Aware Dynamic Pricing algorithms

Deep Learning Research Intern

August 2025 - December 2025

LIPADE, Université Paris Cité, Paris, France

- Researched and implemented self-supervised learning methods for Satellite Image Time Series (SITS) and videos
- Built end-to-end training pipeline for representation learning and temporal analysis of spatio-temporal data using Pytorch and Jean Zay supercalculator

- Benchmarked models against state-of-the-art baselines

Operational Research Intern

February 2025 - July 2025

Société Nationale des Chemins de fer Français (SNCF), Paris, France

- Developed heuristics and metaheuristics for real-time railway traffic management and conflict resolution
- Fixed and implemented new features for real-time railway traffic simulator
- Designed and tested an optimization pipeline in Python, reducing scheduling delays in simulation scenarios

Machine Learning/AI Intern

January 2024 - July 2024

Swiss Life, Zürich, Switzerland

- Contributed to company-wide AI Transformation initiative, deployed ML classifiers for customer feedback categorization and sentiment analysis
- Conducted side research project on exact solutions for Liquid-Time-Constant neural networks with piecewise linear signals
- Wrote research paper as first author accepted for publication (IEEE)

Machine Learning Research Intern

June 2023 - December 2023

Callyope, Paris, France

- Designed and implemented end-to-end speech-processing pipeline secured with Fully Homomorphic Encryption (FHE) and compatible with quantization schemes
- Benchmarked CKKS vs TFHE schemes
- Created quantized methods for audio features extraction
- Contributes to the redaction of a research article and a blog post about Gammatone features

Tutor

2021 - Present

- Tutored mathematics, computer science, physics, and French for high school students preparing for the *baccalauréat*

Skills

- Programming languages: Python, C++, Matlab, Julia
- Software / Tools: GitHub, Google Cloud, Azure ML, TensorFlow, PyTorch, Jean-Zay supercomputer (CNRS/IDRIS)
- Languages: French (native), English (fluent, TOEIC 880), German (intermediate, DSD B1)

Hobbies

- Sport: Running, climbing, swimming, hiking
- Art and creativity : knitting, crocheting, baking/cooking, reading